

In this page, I investigate some typical finite groups.

Specifically, investigate:

Abelian and Abelianization

Commutator and Derived subgroup

Upper-Central Series and Solvability and Composition Series

Center and Conjugacy classes.

Explicit all subgroup

The general properties of S_n , D_n , C_n will be covered in other page.

$\mathbb{Z}_2 \times S_3$, Conjugacy classes.

Since $|G:G_0| = |C_0|$, find the Centralizer of $(0, (12))$.

Since $(1,2)$ commutes only id and itself, and $\mathbb{Z}_2$ is abelian,

$$C_G(0, (1,2)) = \{(0, (1,2)), (0, \text{id}), (1, (1,2)), (1, \text{id})\}.$$

Therefore, the conjugacy classes of $(0, (1,2))$ has $2 \cdot 6/4 = 3$ elements.

Since $\mathbb{Z}_2$ is abelian (or 0 is identity), the first term never change,

and since the conjugate cannot change the cycle structure,

the conjugacy classes of $(0, (1,2))$ is

$$\{(0, (1,2)), (0, (1,3)), (0, (2,3))\},$$

and similarly, the conjugacy classes of $(1, (1,2))$ is

$$\{(1, (1,2)), (1, (1,3)), (1, (2,3))\},$$

$$\text{" } (0, (1,23)) \text{ is}$$

$$\{(0, (1,23)), (0, (1,32))\}$$

$$\text{" } (1, (1,23)) \text{ is}$$

$$\{(1, (1,23)), (1, (1,32))\}$$

and $(0, \text{id})$ and $(1, \text{id})$ have conjugacy classes only containing itself.

$S_3 \times S_3$, Congruency classes.

Let $(1\ 2) \times (1\ 2) \in S_3 \times S_3$, $(1\ 2)$ commute only id or itself, thus

$$C_G((1\ 2)) = \{(\text{id}, \text{id}), (\text{id}, (1\ 2)), ((1\ 2), \text{id}), ((1\ 2), (1\ 2))\}.$$

So, the congruency classes of $(1\ 2) \times (1\ 2)$ has 4 elements

$\sqrt{S_4}$.

$$\begin{array}{l|l} S_4 = & \{ \text{id}, \\ 1,1,2 & (1\ 2), (1\ 3), (1\ 4), (2\ 3), (2\ 4), (3\ 4), \\ 2,2 & (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3), \\ 1,3 & (1\ 2\ 3), (1\ 3\ 2), (1\ 3\ 4), (1\ 4\ 3), (1\ 2\ 4), (1\ 4\ 2), (2\ 3\ 4), (2\ 4\ 3), \\ 4 & (1\ 2\ 3\ 4), (1\ 2\ 4\ 3), (1\ 3\ 2\ 4), (1\ 3\ 4\ 2), (1\ 4\ 2\ 3), (1\ 4\ 3\ 2) \} \\ \text{Partition} & \end{array}$$

A4. Conjugacy classes.

$$A_4 = \{id, (12)(34), (13)(24), (14)(23), \\ (123), (132), (124), (142), (134), (143), \\ (234), (243)\}$$

Since the conjugate cannot change the cycle type, the conjugacy class of each element in A_4 are contained the set of same cycle type. But we don't know it exactly equals to the set, let's calculate.

$$(123)(12)(34)(123)^{-1} = (23)(14)$$

$$(124)(12)(34)(124)^{-1} = (24)(31)$$

therefore, the conjugacy classes of $(12)(34)$ equals the set of $(2,2)$ cycles.

$$(132)(123)(132)^{-1} = (312) = (123) \quad (\text{In actually, } (123)^{-1} = (132))$$

$$(124)(123)(124)^{-1} = (243)$$

$$(142)(123)(142)^{-1} = (413) = (134)$$

$$(134)(123)(134)^{-1} = (324) = (243)$$

$$(143)(123)(143)^{-1} = (421) = (142)$$

$$(234)(123)(234)^{-1} = (134)$$

$$(243)(123)(243)^{-1} = (142)$$

$$(12)(34)(123)(12)(34)^{-1} = (214)$$

$$(13)(24)(123)(13)(24)^{-1} = (341) = (134)$$

$$(14)(23)(123)(14)(23)^{-1} = (432) = (243)$$

therefore, the conjugacy classes of (123) has just four elements.

Generally, since $|G:G_a| = C_a$ and $G_a = C_G(a)$ in the conjugate action, therefore number of elements of the orbit is $|G|/|C_G(a)|$.

Since $C_{A_4}((123)) = \{id, (123), (132)\},$

the orbit of (123) has $|2/3| = 4$ elements

A_5 . The smallest group of non-abelian simple group.

Klein 4-group.

Let $G = \mathbb{Z}_2 \times \mathbb{Z}_2$ be the Klein 4-group.

Then, a map $\varphi: G \rightarrow S_4$:

$$(0,0) \mapsto \text{id}, \quad (1,0) \mapsto (12)(34), \quad (0,1) \mapsto (13)(24), \quad (1,1) \mapsto (14)(23)$$

is Embedding